

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Electronics and Electrical Communication Engineering

2. **Subject Name:** Algorithms

**L-T-P:** 3-0-0

**Credits:** 3

3. **Pre-requisites:** Programming and Data Structures (CS10003)

4. **Syllabus and reference books:**

### Syllabus:

Arithmetic, Fibonacci numbers, Asymptotic notations; Divide and conquer: Multiplication, Recurrence relation and Master theorem, Mergesort, Median, Matrix multiplication, Polynomial Multiplication, Fast Fourier transform; Graphs: Introduction to graphs, directed acyclic graph (DAG), depth-first search, breadth-first search, Dijkstra's algorithm, Bellman-Ford algorithm, shortest path in DAG; Greedy algorithms: Minimum spanning tree, Kruskal's algorithm, Prim's algorithms, Set cover, Huffman encoding; Dynamic programming: Shortest path, Longest increasing subsequences, Edit distance, Knapsack, Floyd-Warshall algorithm, Chain matrix multiplication, Travelling salesman problem; Linear programming: Duality, Simplex algorithm, Flows in networks, Bipartite matching, Zero-sum games; NP-complete problems: Search problems, P and NP, Backtracking, Branch and Bound, Approximation algorithms, Heuristics; Randomized algorithms: Las Vegas, Monte Carlo, Hashing.

### Reference Books:

- 1) Sanjoy Dasgupta, Christos Papadimitriou, Umesh Vazirani, Algorithms, McGraw-Hill Education, 2006.
- 2) Thomas Cormen, Charles Leiserson, Ronald L. Rivest and Clifford Stein, Introduction to Algorithms, MIT Press, 2009.
- 3) Jon Kleinberg and Éva Tardos, Algorithm Design, Pearson, 2013.
- 4) Udi Manber, Introduction to Algorithms: A Creative Approach, Addison-Wesley, 1989.

### 5. Lecture-wise break-up:

| Sl. No. | Topic                                                                                                                                                                   | No. of lectures |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.      | Arithmetic, Fibonacci numbers, Asymptotic notations                                                                                                                     | 3               |
| 2.      | Divide and conquer: Multiplication, Recurrence relation and Master theorem, Mergesort, Median, Matrix multiplication, Polynomial Multiplication, Fast Fourier transform | 6               |

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|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>3.</b>                    | Graphs: Introduction to graphs, directed acyclic graph (DAG), depth-first search, breadth-first search, Dijkstra's algorithm, Bellman-Ford algorithm, shortest path in DAG       | <b>7</b>  |
| <b>4.</b>                    | Greedy algorithms: Minimum spanning tree, Kruskal's algorithm, Prim's algorithms, Set cover, Huffman encoding                                                                    | <b>5</b>  |
| <b>5.</b>                    | Dynamic programming: Shortest path, Longest increasing subsequences, Edit distance, Knapsack, Floyd-Warshall algorithm, Chain matrix multiplication, Travelling salesman problem | <b>6</b>  |
| <b>6.</b>                    | Linear programming: Duality, Simplex algorithm, Flows in networks, Bipartite matching, Zero-sum games                                                                            | <b>5</b>  |
| <b>7.</b>                    | NP-complete problems: Search problems, P and NP, Backtracking, Branch and Bound, Approximation algorithms, Heuristics                                                            | <b>3</b>  |
| <b>8.</b>                    | Introduction to Randomized algorithms: Las Vegas, Monte Carlo, Hashing                                                                                                           | <b>3</b>  |
| <b>Total number of hours</b> |                                                                                                                                                                                  | <b>38</b> |